Forest Policy and Economics 189 (2026) 103830

Contents lists available at ScienceDirect

Forest Policy and Economics

journal homepage: www.elsevier.com/locate/forpol

Stakeholders' perception about forest fires in the Alpine region

The Trung Hoang [a,*], Lena Vilà-Vilardell [b], Mortimer M. Müller [a], Giorgio Vacchiano [c], Gianni Boris Pezzatti [d], Sylvain Dupire [e], Christian Schunk [f,1], Harald Vacik [a]

[a] Institute of Silviculture, University of Natural Resources and Life Sciences (BOKU), Peter-Jordan-Straße 82/II, 1190 Vienna, Austria
[b] Joint Research Unit CTFC – AGROTECNIO, Ctra de St. Llorenç de Morunys, km 2, 25280 Solsona, Spain
[c] Department of Agricultural and Environmental Sciences (DISAA), University of Milan, Via Celoria 2, 20133 Milan, Italy
[d] Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Insubric Ecosystems Research Group, Research Campus Cadenazzo, 6593 Cadenazzo, Switzerland
[e] Université Grenoble Alpes - INRAE UR LESSEM. Grenoble, France
[f] Formerly: Professorship of Ecoclimatology, TUM School of Life Sciences, Technical Universiy of Munich, Hans-Carl-von Carlowitz-Platz 2, 85354 Freising, Germany

ARTICLE INFO

Keywords:
Wildfire governance
Policy actors
Socio-economic factors
Natural drivers
Risk perceptions
Alpine region

ABSTRACT

Climate change has increased wildfire pressure in the European Alps, yet a common understanding of fire regimes and coordinated fire management strategies remains lacking. This study analyzes the perceptions of three stakeholder groups in Austria, France, Germany, Italy, Slovenia, Liechtenstein, and Switzerland. An online survey and a forest fire workshop for experts were conducted to obtain different perceptions regarding the key factors and trends of the current and future fire regime. Our analysis indicates that there are significant differences between the groups regarding the perception of the number of fires, burnt areas, and future trends. Stakeholders identified precipitation, temperature, and wind as the main natural drivers of the fire regime, while recreational activities, rural abandonment, and owner awareness were identified as the main socio-economic drivers. These findings advance our understanding of the complex systems shaping future fire regimes in the European Alps and provide decision-makers with tools to design effective integrated fire management strategies.

1. Introduction

Wildfires are an increasing issue worldwide and are also increasingly becoming a challenge in regions where the fire regime was previously characterized by low-intensity and infrequent wildfires. At the European Union (EU) scale, more than 60,000 forest fires are recorded each year, with on average around half a million hectares burned and associated direct economic losses of approximately EUR 2–3 billion per year (EASAC, 2025; OECD, 2023). Climate change is widely recognized as a dominant macro-driver of these trends, interacting with land-use change to produce larger and more intense fires (EASAC, 2025). One of these regions is the European Alps (Conedera et al., 2018; Vacchiano et al., 2018), traditionally not considered a fire-prone area, which is facing an increased forest fires activity, to which the vegetation is not adapted and which may potentially endanger lives, cause economic damage, and change mountain forests' ecological and protective functions due to a combination of factors (Canepa and Drogo, 2021; Gerberding and Schirpke, 2025; Sevinc et al., 2020; Wastl et al., 2013). As geographical conditions, south-facing slopes dominated by coniferous forests on carbonate rock may experience stand-replacing forest fires due to drier vegetation and a thin humus layer (Müller et al., 2020), while fires tend to spread faster in steeper slopes (Conedera et al., 2018; Csontos and Cseresnyés, 2015; Lahaye et al., 2018; Valese et al., 2014). Moreover, the drivers of fire spread in the Alps are experiencing fast changes (Müller et al., 2020). On one hand, the different emission scenarios let us assume that the average air temperature in the Alpine region will increase from 1 to 4 °C by 2100 (Kotlarski et al., 2023; Wastl et al., 2012). Also, precipitation amounts are expected to decrease in summer with more and prolonged drought periods and longer heat waves (Gobiet et al., 2014).

On the other hand, the Alpine region is facing changes differently across various sub-regions in socio-economic conditions, such as rural abandonment and more recreational activities. Rural abandonment results in greater fuel accumulation but also leads to uncontrolled practices of pastoral burning (Müller et al., 2020; Pezzatti et al., 2013). The increase in recreational activities leads to a higher number of people

* Corresponding author.
E-mail address: trung.hoang@boku.ac.at (T.T. Hoang).
[1] Current work address: Die Autobahn GmbH des Bundes, Niederlassung Nordbayern, Flaschenhofstraße 55, 90402 Nuremberg, Germany.

https://doi.org/10.1016/j.forpol.2026.103830
Received 28 August 2025; Received in revised form 15 May 2026; Accepted 27 May 2026
Available online 4 June 2026
1389-9341/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

visiting the forest, potentially increasing the number of fire ignitions (Vacchiano et al., 2018). All these factors will cause greater damage and ecological stress, probably also in protective forests, increasing the vulnerability to natural hazards, slowing down the recovery of forest vegetation and its services, and resulting in higher costs for fire suppression and restoration measures.

Moreover, national-level data compiled by the OECD (2023) indicates that public expenditures for wildfire management have increased substantially in several European countries, with the largest share commonly going to suppression rather than prevention. Such cost dynamics propagate to broader forest management budgets, since wildfires generate post-fire restoration and reforestation needs, alter land-use trajectories, and affect the long-term provision of timber and protective ecosystem services (Rego et al., 2013; OECD, 2023). EASAC (2025) also highlights this imbalance and recommends a reallocation towards proactive, landscape-level prevention measures and integrated land-use planning. A multi-level legal and policy framework on wildfire risk has emerged in Europe. At the EU level, wildfire-related provisions are embedded in a fragmented set of sectoral instruments rather than in a dedicated framework directive (Galiana et al., 2013; Montiel-Molina, 2013; EASAC, 2025). This fragmentation is consistent with recent literature on wildfire governance, which argues for a shift from suppression-centered approaches towards proactive, people-centered, and integrated fire management across prevention, preparedness, response, and recovery (Wunder et al., 2021). These instruments include the EU Forest Strategy for 2030, the EU Biodiversity Strategy for 2030, the European Green Deal and the EU Strategy on Adaptation to Climate Change (COM (2021) 82 final), and operational mechanisms such as the EU Civil Protection Mechanism and the Common Agricultural Policy (EASAC, 2025; OECD, 2023). towards an integrated framework linking forest resilience, biodiversity protection, climate adaptation, civil protection and landscape-level prevention. The EU Forest Strategy for 2030 emphasizes forest protection, restoration, and climate resilience, while the EU Biodiversity Strategy for 2030 and the Nature Restoration Regulation connect ecosystem recovery with resilience to climate-related hazards, including forest fires. The EU Strategy on Adaptation to Climate Change further promotes systemic adaptation, and EFFIS, supported by the Expert Group on Forest Fires, provides harmonized wildfire monitoring and risk information for Europe. At the response level, the EU Civil Protection Mechanism and rescEU strengthen cross-border firefighting capacity, while the Alpine Convention and its Mountain Forests Protocol provide a region-specific framework for sustainable mountain-forest management and fire prevention. Together, these instruments increasingly emphasize integrated wildfire risk management across prevention, preparedness, response, and recovery (EASAC, 2025).

Several research projects have studied the occurrence, distribution, causes, and characteristics of forest fires in the Alpine region (Conedera et al., 2018; Curt et al., 2016; Dupire et al., 2017; Vacik et al., 2011; Zumbrunnen et al., 2012). Case studies related to fuel management and fire behavior modeling have also been conducted in the Alpine region (Arpaci et al., 2013; Rizzolo, 2016). Although lightning fires may be relevant during summer with prolonged drought, most forest fires in the European Alps are a result of anthropogenic influence (Müller et al., 2020). The fires may occur due to negligence, arson, accidental ignitions, and forest and agricultural practices such as discarded cigarettes, cleaning works, hot ashes, sparks from railroads or power lines (Conedera et al., 2006; Müller et al., 2013; Vacik et al., 2011). Recent studies have indicated that differences between regions in prioritizing fire prevention and suppression measures, and fuel management have been linked to variations in burnt area size (Carvalho et al., 2010). More research is needed to understand the key drivers of the fire regime and stakeholders' perceptions and perspectives on fire management. Understanding stakeholders' preconceptions and motivations is critical for designing appropriate fire prevention measures, improving their effectiveness, and ensuring that those measures are accepted, shared, and implemented consistently across organizations and communities (Costa and Menichini, 2013; Dwivedi et al., 2016; Tedim et al., 2022).

However, there is a lack of studies focusing on the stakeholders' perceptions of forest fires in the Alpine region. It is assumed that stakeholders' and communities' knowledge about forest fires is limited. More emphasis must be placed on awareness-raising activities that include all relevant stakeholders, as the lack of participation by households in the wildland-urban interface or mountainous areas and differences in education and jobs among respondents are regularly pointed out as barriers to fire management activities (Aguilar and Montiel, 2011; Müller et al., 2020). Different stakeholder groups have their own socio-environmental values and perspectives on the ecological value and challenges of forest fires which can lead to conflicting situations (Castillo et al., 2021; Grilli et al., 2016; Molina-Terrén et al., 2016; Paletto et al., 2013). This is especially the case when empirical data are lacking (Pereira et al., 2016). For example, there is a difference in perception of the ecological benefits of fire across different job roles. While forest rangers placed a lower importance on this factor, groups such as researchers and fire chiefs rated it as very important. When it comes to the intentional use of fire, prescribed burning for prevention faces issues like limited social acceptance and bureaucratic hurdles. In contrast, using fire during active suppression usually involves fewer obstacles (Molina-Terrén et al., 2016).

Additionally, the knowledge of how different stakeholders perceive various natural and socio-economic factors influencing forest fire regime will deepen our understanding of their work and actions towards forest fires. Taken all together, the aims of this study are to (1) understand the current stakeholders' perceptions regarding the trend of forest fire occurrence in Austria, France, Germany, Italy, Slovenia, Liechtenstein, and Switzerland and to (2) investigate the differences in the stakeholders' perceptions on natural and socio-economic driving factors of the forest fire regime. The outcomes of this study will help to raise awareness on the knowledge gaps in forest fire management and will provide an overview of the challenges and obstacles for managers and decision-makers to improve their fire management strategies and promote stakeholders' participation.

2. Materials and methods

2.1. Study area

The study area covers the Alpine territories of seven countries (Fig. 1).

2.2. Data collection

Stakeholders' perceptions on forest fire occurrence, management strategies, and natural and socio-economic factors driving the fire regime in the study region were collected through a survey and a workshop with forest fire experts of three stakeholder groups (authorities, scientists, and action forces, Table 1). The multilingual online survey was conducted in 2019, in the context of the EUSALP study "Forest fires in the Alps: State of knowledge and future challenges" (Müller et al., 2020). The survey was distributed in multiple languages (at minimum German, French, Italian, and Slovenian, reflecting the language communities of the study region) via the network of fire experts of the Global Fire Monitoring Center (GFMC).

The survey has provided first insights about the fire management practices in the Alpine region (Müller et al., 2020). A snowball sampling approach was employed in the survey to contact participants from the three identified stakeholder groups using the network of fire experts of the Global Fire Monitoring Center (GFMC) in the participating countries. The workshop held in Vienna (June 2019) served as a follow-up discussion and contextualization forum that allowed us to discuss the survey findings with participants.

To set a common understanding, a uniform definition of a forest fire

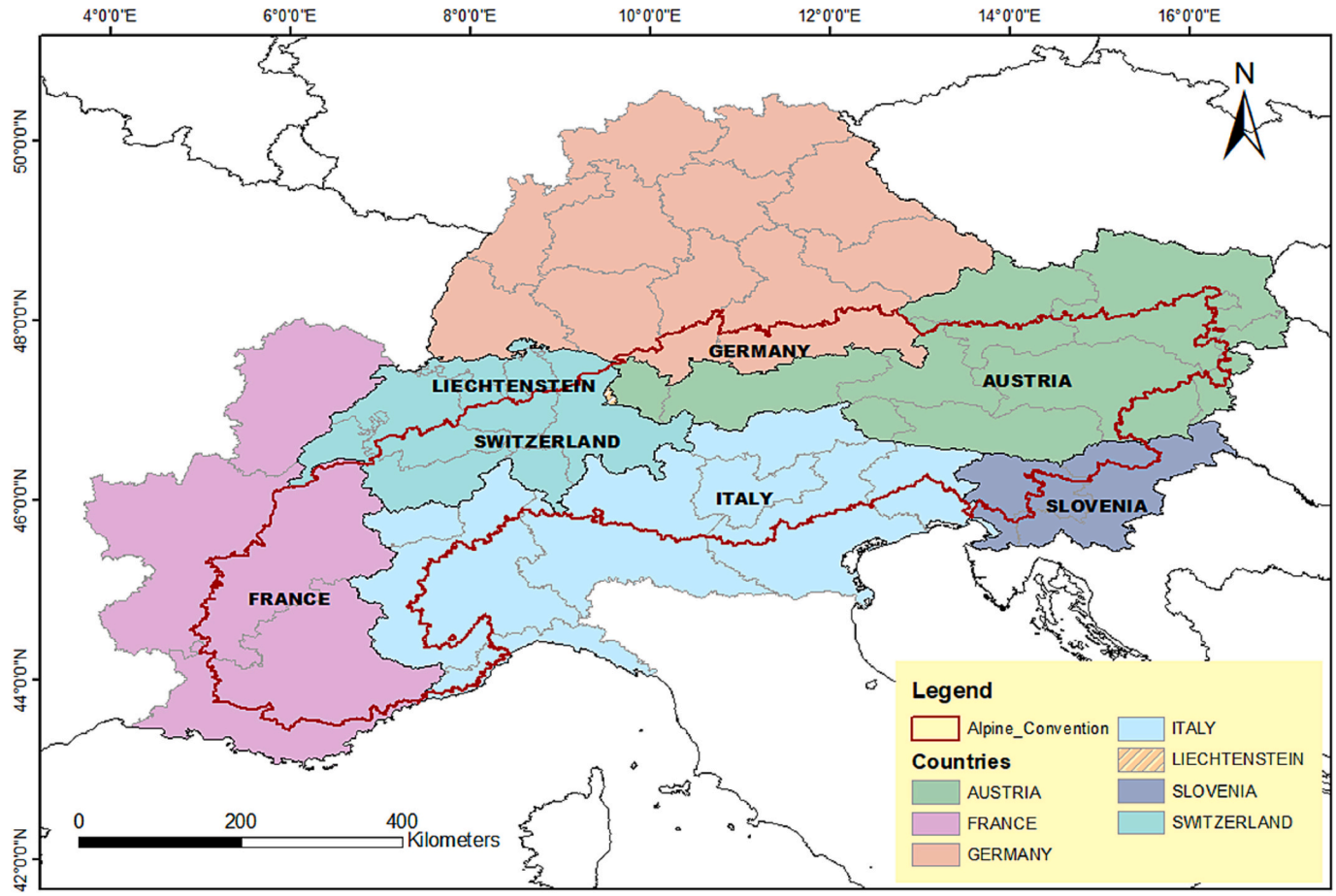


Fig. 1. Map of the study region (red polygon) in the European Alps base on the EU Strategy for the Alpine Region (EUSALP), including Alpine territories of Austria, France, Germany, Italy, Slovenia, Liechtenstein, and Switzerland. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Description of targeted stakeholder groups.

Stakeholder group	Definition
Authorities	Renowned persons in civil service responsible for or mainly engaged in planning and conducting measures for forest fire prevention or post-fire restoration measures on a transregional or nationwide basis (e.g., members of forest authorities, members of advisory services, operators of early warning systems).
Scientists	Renowned persons with a scientific background in fire research or active in forest fire monitoring and modeling on a nationwide or pan-European basis.
Action Forces	Renowned persons from fire brigades, aerial firefighting, police, or national army responsible for or with a mainly focused on forest fire fighting on a transregional or nationwide basis.

was established at the beginning of the survey: "A forest fire is an uncontrolled fire in forested area (including clearcuttings, young forest, coppice and vegetation at the alpine forest border), independent of fire type (smoldering fire, surface fire, crown fire), size and cause (e.g., also burning tree from lightning).". In this study, we have also conceptualized wildfire governance-relevant perceptions as actors' subjective, policy-relevant assessments of wildfire dynamics that are sufficiently specific to be operationalized and compared across stakeholder groups. This approach allowed us to analyze wildfire risk perceptions in relation to governance and policy support (Alló and Loureiro, 2020).

The questionnaire contained three main parts with eight sections and 61 questions in total (see the supplementary material files).

The first part of the questionnaire aimed to identify the assumed trends of the forest fire regime in the study region. The questions focused on stakeholders' perceptions of the fire situation as well as the importance of natural and socio-economic factors as drivers of the fire regime. The second part of the questionnaire consisted of three sections, namely fire prevention, forest management and firefighting. The last part of the questionnaire assessed the participants' perceptions of the role of political frameworks and risk governance. Stakeholders were also able to express their preferences regarding forest fire management planning and the implementation of fire prevention measures at the local to national level. In total, there were 80 respondents to the survey. Details of the respondents are presented in Table 2.

Table 2
Demographic characteristics of respondents.

	Action Forces	Authorities	Scientists	Total
Austria	8	11	4	**23**
Switzerland	2	17	3	**22**
Germany	2	3	2	**7**
France	2	2	6	**10**
Italy	3	3	5	**11**
Liechtenstein	–	1	–	**1**
Slovenia	–	5	1	**6**
Total	**17**	**42**	**21**	**80**

To discuss the outcomes of the survey, a workshop with forest fire experts was held in Vienna with scientists and representatives of authorities and action forces from all Alpine countries.

2.3. Data analysis

Comparison of stakeholders' perceptions regarding the number of forest fires and burnt areas was carried out by analyzing the responses received to the corresponding questions in the survey. Only questions and responses related to the two above-mentioned aims of this present research were used for data analysis. Differences between the responses of stakeholder groups were examined using Kruskal-Wallis test (Kruskal and Wallis, 1952). If significant differences were identified, Dunn's post-hoc test with the Hochberg method (Dunn, 1964; Hochberg and Tamhane, 1987) was conducted to clarify differences within stakeholder groups. Additionally, we used principal component analysis (PCA) to analyze combined responses to several questions, aiming to identify any patterns or associations among the respondent groups. Detailed plots of the PCA results are included in the Supplementary Material (Figs. S4 and S5). The statistical analysis was done in R software version 4.4.1 (R Core Team, 2024).

3. Results

3.1. Perceptions of the number of forest fires and burnt areas

Most stakeholders indicated an increase in the number of fires in the future, with the group of action forces having the highest share of answers pointing to a strong increase. None of the groups expected a strong decrease of fire risk in the next ten years. Scientists mostly chose either decrease, increase, or constant trends. Authorities had the most diverse answers (from decrease to strong increase) (Fig. 2).

Answers by the action forces and scientists regarding the number of fires showed statistically significant differences in the perceptions of the two stakeholder groups ($p < 0.05$). The scientist group expected a significantly lower increase of the number of forest fires in the study region than representatives of action forces.

Most scientists, authorities, and action forces expected an increase of the burnt areas in the next ten years. The action force group again had the highest share for answers, pointing to a strong increase. No participant in the survey expected the size of the burnt area to strongly decrease. However, scientists' responses included all four options from decrease to strong increase. Representatives of authorities did not assume a strong increase of the burnt areas. Their responses ranged from a decrease to an increase. Action forces chose only two response options (increase and strong increase, cf. Fig. 2).

Statistically significant differences in the expected trend of burnt areas could be found between the action forces group and the two other groups ($p < 0.05$). Members of action forces in all Alpine countries believe that the burnt area will be significantly higher than what was expected by the other groups.

Additionally, action forces are situated on the upper-right quadrant of the plot in Fig. 3, corresponding to a "strong increase" in fire occurrence and burnt area. Conversely, scientists are shown a more moderate perception overall, centered around increases in both variables. While there is some overlap in ellipses, the spatial distribution of points indicates that action forces predicted a more significant increase in fire frequency compared to other groups.

3.2. Comparing actual data with perceptions of burned areas and number of fire predictions

Within our sample in Austria, Switzerland, Germany, and Slovenia, most participants tend to choose a class with fewer than one hundred fires per year. In contrast, a significant number of participants in France and Italy consistently select more than one thousand fires annually.

Most respondents among scientists, authorities, and action forces in Austria, Germany, and Switzerland assumed that the mean size of an Alpine forest fire is less than one hectare. In France, Italy and Slovenia the mean size of a forest fire in the Alps is believed to be 1 to 10 ha (Fig. 4).

According to the EFFIS dataset, on average, the annual number of fires in Austria, France, Slovenia, and Switzerland did not exceed 250, indicating that the limited participation from these countries provided results that closely align with the actual statistical data. Regarding the average annual single fire burns less than 0.1 ha in Austria, Germany, and Switzerland, 0.5 ha in Slovenia, and 1 ha in France and Italy. The majority of stakeholder perceptions related to the burned area across the study areas in Austria, Germany, Switzerland, France, and Italy showed, therefore, a low accuracy on this parameter, as most of the stakeholders assumed a higher mean burned area compared to reality, the EFFIS reference (2006–2017).

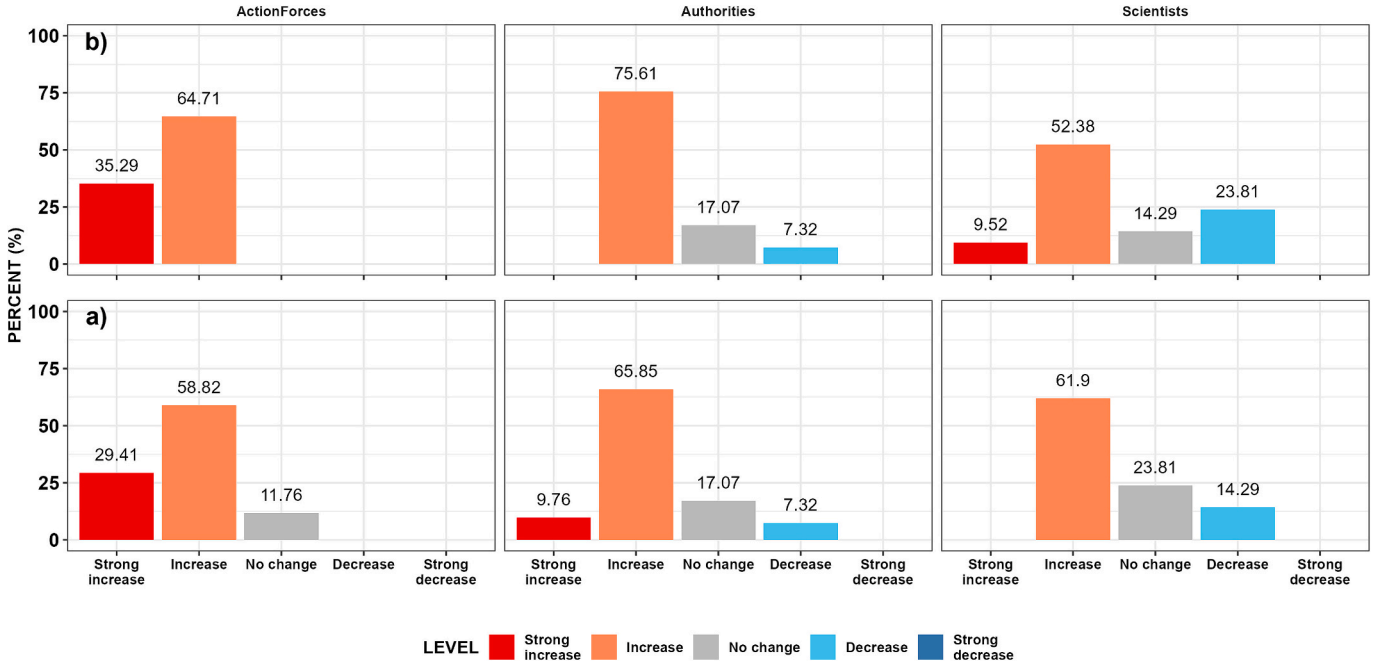


Fig. 2. Expectations regarding the (a) number of forest fires and (b) burnt areas among the three stakeholder groups.

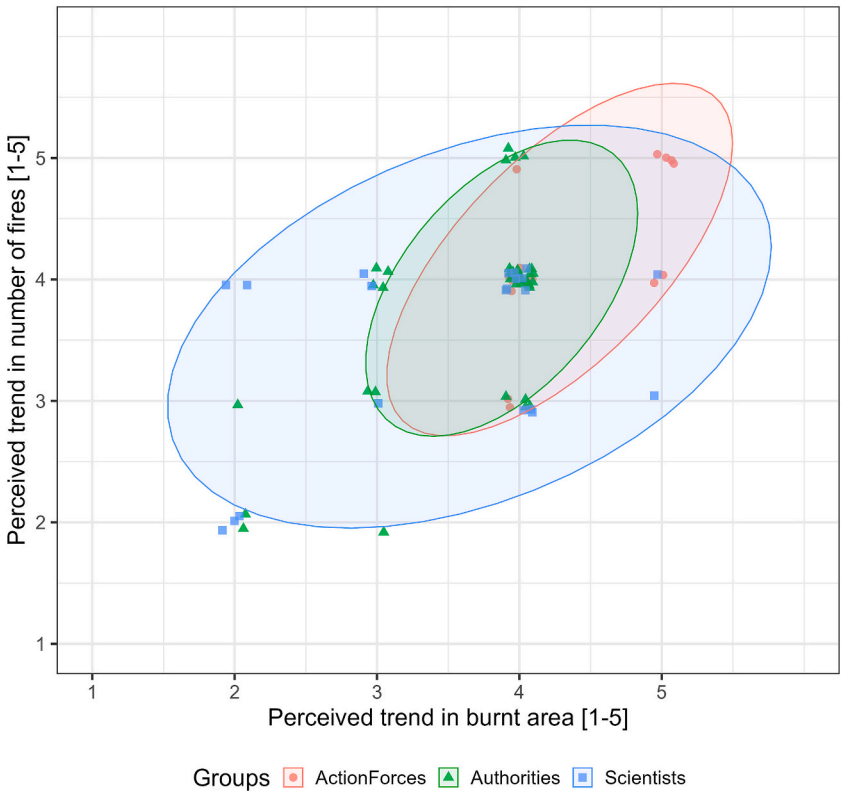


Fig. 3. Perceived trends in the number of forest fires and burnt areas as assessed by three stakeholder groups. Rating scale: 1 = strong decrease, 2 = decrease, 3 = no change, 4 = increase, 5 = strong increase.

3.3. Natural factors affecting fire regime

Fig. 5 reports the stakeholders' ratings for the question about the importance of the different natural factors as drivers for the present and future fire regime in the studied countries.

According to the stakeholders' perceptions, "precipitation", "wind", and "temperature" were identified as the most important natural factors influencing the fire regime, both in the current and the future trends.

The meteorological variables were followed by "fuel quality in forests" (i.e., moisture, structure, and compactness) and "fuel load in forests", with the highest ranking given by the action forces group, followed by the scientists and authorities group in the present and future. "atmospheric stability" was ranked lowest among the six natural factors, with the lowest ranking from authorities. Additionally, the importance of "fuel quality in forests" and "fuel load in forests" were nearly the same for all three groups for present and future trends. These factors were ranked the most important level affects the fire regime by action forces group, then by the remaining groups. Overall, all stakeholder groups agreed that the six natural factors will gain importance in the future.

3.4. Socio-economic factors affecting the fire regime

Fig. 6 illustrates the importance of the eight socio-economic factors as drivers for the present and future fire regime in the stakeholders' country.

All stakeholder groups rated "recreational activities", "rural abandonment" and "owner awareness" as the most important socio-economic drivers for both the past and present fire regime. "Intentional fires" are considered the least important socio-economic factor. "Recreational activities" had nearly the same importance for all three stakeholder groups. The action forces and scientists expressed similar perceptions on the importance of most factors, while the authorities had varying levels of importance.

Authorities assumed that "change in forest management", "rural abandonment", "owner awareness", and "administrative reforms" are less important, whereas members of action forces considered these factors as more important, especially "rural abandonment".

In summary, in the present, scientists associate fire risk with factors such as high fuel load, precipitation, wind, and temperature, highlighting the importance of these factors in shaping the fire regimes. Action forces linked fire risk to moderate levels of atmospheric instability and fuel quality factors. Authorities assigned a lower influence to "atmospheric stability", "temperature", and "fuel load".

In the current and future timeframe, action forces seem to be more inclined to assign responsibility to high fuel loads and quality, as well as atmospheric conditions. Scientists emphasized the role of high temperatures and wind, highlighting the crucial role of climate-related factors in shaping future fire risks. In contrast, authorities tended to assign medium to low importance to variables such as temperature, fuel quality, and atmospheric conditions.

Scientists have identified several key factors that influence current fire regimes, including "intentional ignitions", "rural abandonment" with high importance, and "infrastructures" and "recreational activities" with moderate importance. Action forces assigned a moderate importance to "rural abandonment", "administrative", and "change in forest management". Authorities assigned a lower importance to recreational and intentional factors. For the future, action forces seem to pay more attention to the role of administrative and forest management.

4. Discussion

4.1. Perceptions of the number of forest fires

Most stakeholders believed that there would be an increase in the number of forest fires in the studied countries of the Alpine region. As a

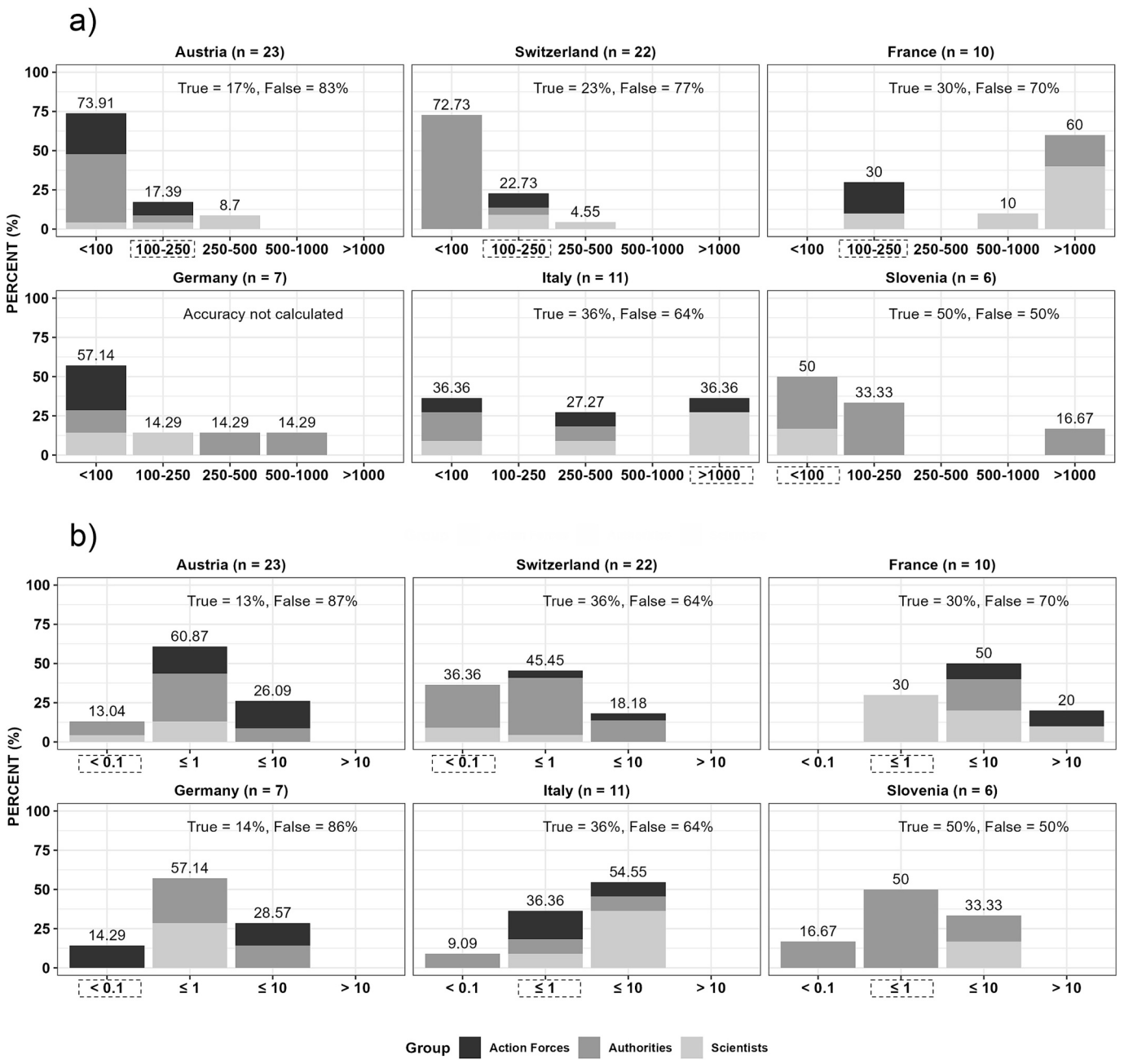


Fig. 4. Stakeholder groups and assumptions regarding the (a) number of forest fires and (b) sizes of burnt areas from individual forest fires in Alpine countries (in %). The dashed box on the chart indicates the reference value. The EFFIS (European Forest Fire Information System) dataset was used with the mean annual number of fires and the mean annual burned area in the Alpine region (Austria, France, Germany, Italy, Slovenia, and Switzerland) from 2006 to 2017. No data available for Liechtenstein information for the study area and period.

result of climate change, the coming decades will likely face longer and more intense drought periods and heat waves, increasing a higher fire hazard, as a result of rural abandonment and recreational activities (Gobiet et al., 2014; Müller and Vacik, 2017; Trnka et al., 2016). Additionally, long-term fire suppression and selective logging have unintentionally created denser, more homogeneous forest structures and increased fuel loads. Unwanted consequences from these activities in the past support stakeholders' concerns regarding a potential rise in the frequency of forest fire occurrences (Merschel et al., 2014; Naficy et al., 2010). While the leading cause of forest fires in the European Alps is negligence, there is also an expected increase in the number of lightning-caused fires with climate change (Conedera et al., 2006; Müller et al., 2020; Vacchiano et al., 2018).

Scientists estimated the increasing trend of the number of fires to be significantly weaker compared to the action forces group. Scientists tend to make cautious forecasts by evaluating different scenarios, while action forces may be more biased by recent fire events. The perceptions of each stakeholder group depend on their background and experiences in different regions and under various fire conditions. The findings are consistent with studies investigating long-term trends in meteorological forest fire danger across different regions of the Alps. The significant warming, combined with decreasing humidity and changes in precipitation patterns in the Southern and Western Alps, leads to a strong increase in forest fire danger. In contrast, the Northern Alps have only experienced a slight increase in the number of days with high fire risk. Furthermore, perceived differences between groups can be explained by

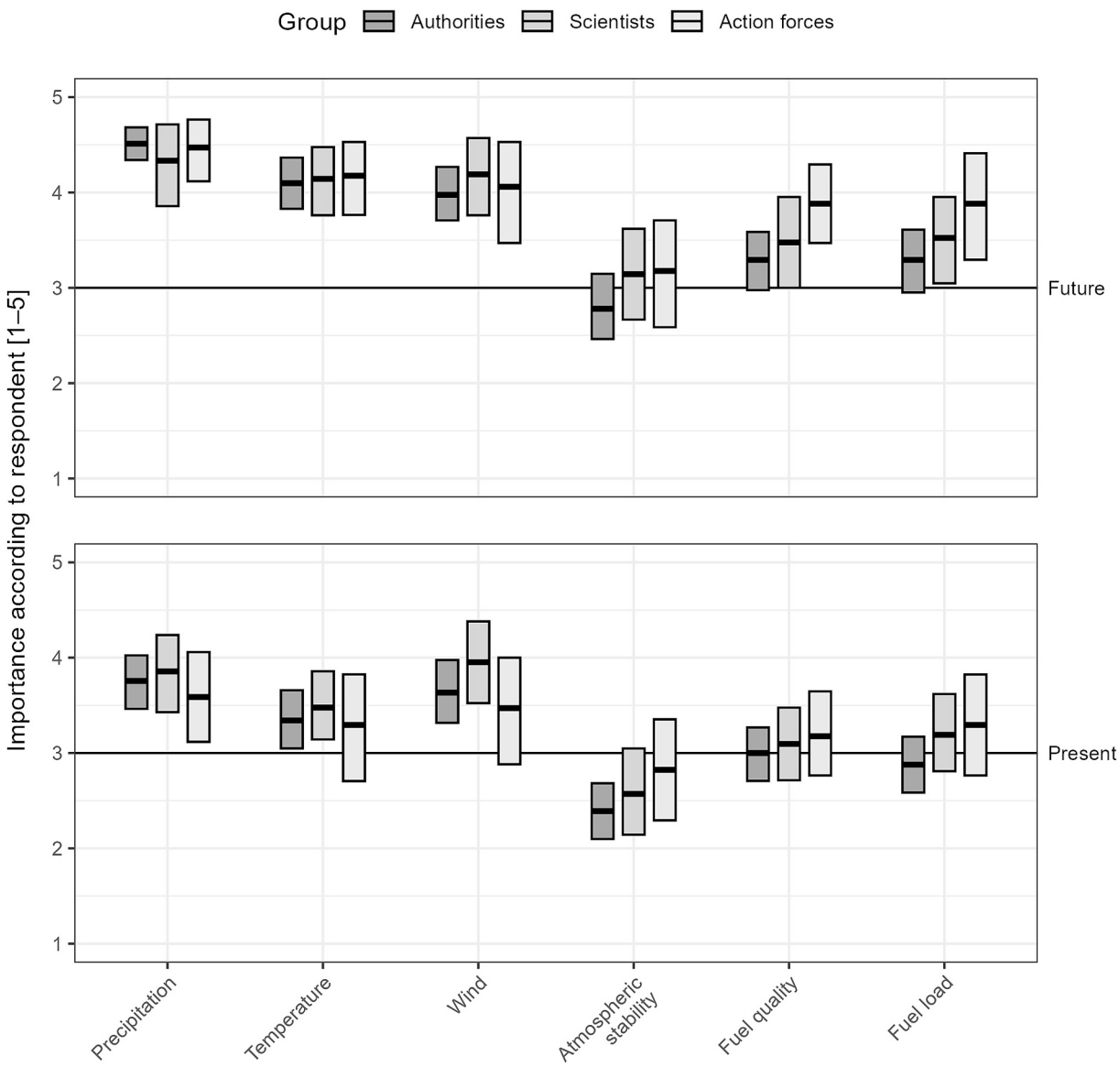


Fig. 5. Crossbar plot of the average rank of the importance of the six natural factors as drivers for the present and future fire regime in the study region. Rating scale: 1 = very low, 2 = low, 3 = medium, 4 = high, 5 = very high.

the fact that when authorities face complex choices in decision-making, they may consider more "safe" options that are assumed to be less risky. This fact is also called "certainty bias" (Kahneman and Tversky, 1979; Maguire and Albright, 2005). Studies have shown that respondents often underestimate risks that they have never been exposed to and overestimate risks they deal with more often (Kahneman and Tversky, 1979; Slovic et al., 1982). Other studies showed that experts estimate fire risk more comprehensively compared to the general public because they consider scientific evidence (Slovic, 1987; Swim et al., 2011).

It is also important to mention that a biased perception of the causes of forest fires can significantly influence stakeholders' perceptions on fire frequency. Negligence and human activities are often cited as the most prevalent causes, and preventive measures or changes in the legal framework can have a huge impact on the occurrence of fires. For instance, a study conducted in Ticino, Switzerland, revealed that the number of forest fires was reduced by over 50% in a few years, following an air quality law that prohibited the burning of garden debris in the open (Pezzatti et al., 2013). Additionally, cultural background plays a vital role in fire use and exposure norms. In the northern Alps, open-fire activities like grilling are common, while in the southern Alps, they are limited due to a higher awareness of wildfire risks. These differences highlight the need for prevention strategies that blend legal measures with cultural understanding (Curt et al., 2016). Hence, while human-caused ignition is the most likely trigger for fires, tailored preventive measures can significantly reduce ignition opportunities, which highlights the need for an unbiased perception of the major fire causes.

4.2. *Perceptions of the burnt area*

All stakeholder groups expected that there would be an increase in the burnt areas in the next ten years. This is consistent with studies from recent years which showed that current fire seasons are shifting and expanding, or even new fire seasons are developing (Müller et al., 2015; Wastl et al., 2012). Especially late autumn and winter fires may become more relevant in the future, as the years 2011, 2015, and 2018 have already shown (Müller et al., 2020; Schunk et al., 2013). Increased burned areas are also expected not only in the south of the French Alps but also at the lower altitude in the north as a result of more extreme droughts (Dupire et al., 2019).

The differences among the stakeholder groups could have been

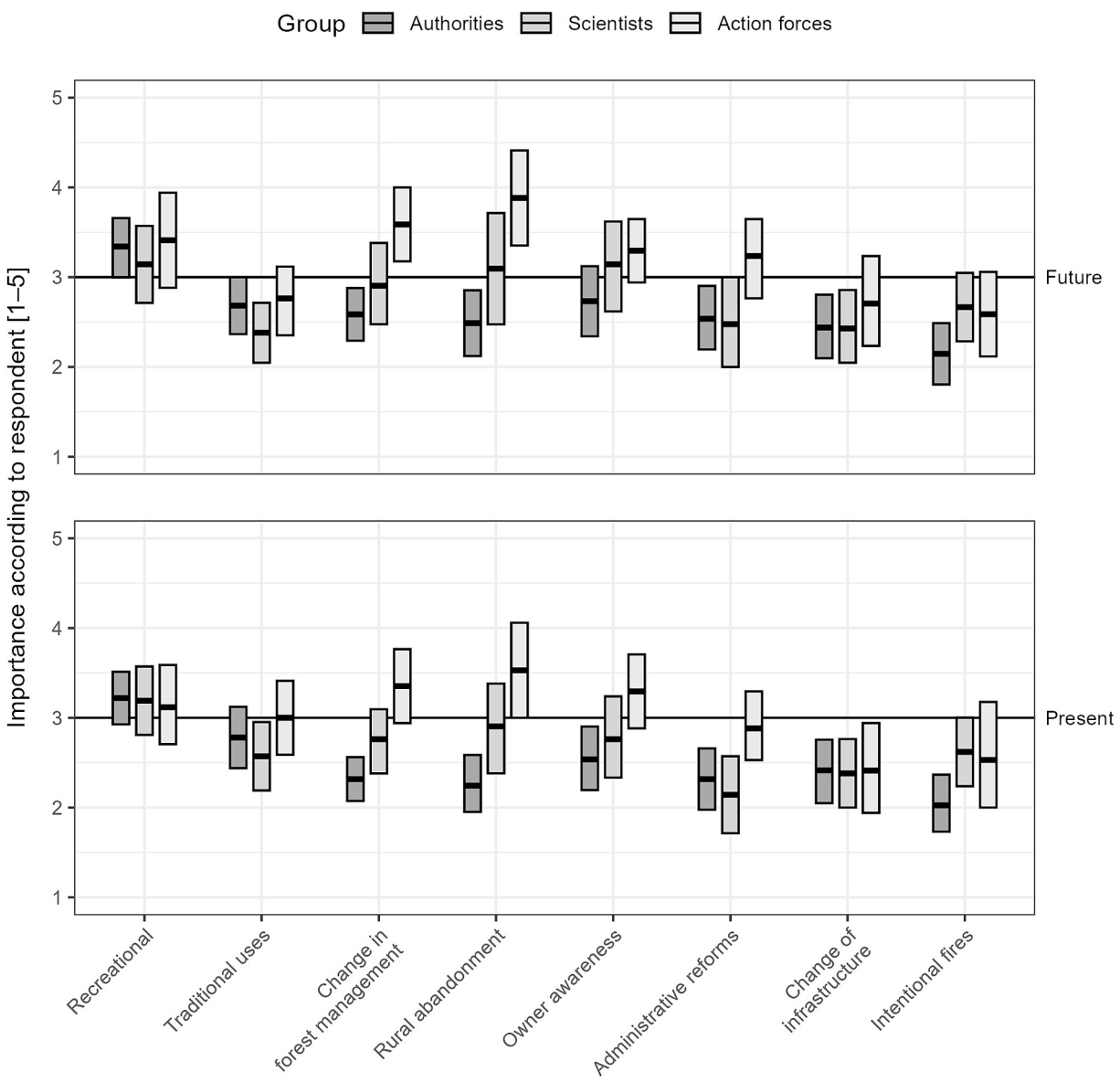


Fig. 6. Crossbar plot of the average rank of the importance of the socio-economic factors as drivers for the present and future fire regime in the study region. Rating scale: 1 = very low, 2 = low, 3 = medium, 4 = high, 5 = very high.

explained by the varying experiences on the field, the influence of the media on climate change, and news reports about extreme forest fire seasons in Brazil, the U.S, and Australia. Previous studies have demonstrated that there is a strong diversity of perceptions regarding forest management activities and forest-related processes within and between stakeholder groups which are often driven by the growing awareness of the complexity and mutual interdependence of such environmental problems (Hickey and Innes, 2005; Raftoyannis et al., 2014).

Predicting forest fire trends is complex, needs to consider different criteria, and is difficult to judge based on observations (Raftoyannis et al., 2014). During the forest fire workshop in Vienna in 2019, it was evident that the perception of the annual number of forest fires and burnt areas differed considerably between action forces and authorities. Authorities may also have more accurate perceptions because they work with real fire data to implement forest fire management strategies. Researchers have more opportunities to gain access to information and to studies from other countries. Also, action forces often rely more on their own experiences in the field (Raftoyannis et al., 2014).

This may also lead to heterogeneous perceptions regarding the burnt areas. To adapt the different experiences and approaches, collaborations among stakeholders are necessary, not only within scientific studies but also through transnational trainings and specific forest fire scenarios for fire brigades and action forces.

Studies indicated that reducing the area burned by wildfires is significantly more complex than simply lowering the number of ignitions. It necessitates mid- to long-term investments in preparedness and capacity-building. These investments include organizing and training fire brigades, implementing prescribed burning practices, and enhancing suppression infrastructure, such as establishing water points for helicopters (Fernandes and Botelho, 2003; Pezzatti et al., 2013). Additionally, in Southern Europe, even with established integrated fire management systems, it is vital to continually adapt to shifting fire patterns caused by climate change, land abandonment, and emerging ignition sources (Papathoma-Köhle et al., 2025; Tedim et al., 2018). Unfortunately, these complexities are often overlooked by those who are not experts, leading to an underestimation of the long-term resources required for effective fire management (Pandey et al., 2023).

4.3. Perceptions of interacting drivers in the Alpine forest fire regime

Stakeholders in our sample perceive wildfire risk in the studied countries of the Alpine region as an outcome of the interaction of natural and socio-economic drivers, rather than from a single dominant cause. This supports rather an integrated perspective in which weather-related hazards influence the likelihood of fire occurrence and spread, while land-use change and human activities create ignition opportunities and shape the socio-political feasibility of preventive efforts (Conedera et al., 2018). Across groups, respondents identified precipitation, temperature, and wind as key natural drivers of the current and future fire regime, while also highlighting socio-economic pressures such as recreational activities, rural abandonment, and owner awareness as influential for the wildfire risk.

Temperature in the Alpine region has already increased by +1.5 to +2 °C compared to the baseline climate before 1990 and is expected to rise further in the coming decades (Auer et al., 2007; Huppmann et al., 2025). It is also likely that heat waves and drought periods will be longer and more severe in the summer months (Fernandez-Anez et al., 2021; Gobiet et al., 2014; Trnka et al., 2016). The trend towards more drought and heat is expected to lead to more fire incidents, especially in southern European countries (Turco et al., 2018). A similar picture is expected for the Alpine region (Wastl et al., 2012).

The fire regime in the Alps is highly heterogeneous but increasingly driven by the weather conditions (Conedera et al., 2018). Scientists had a tendency to rank wind, temperature, and precipitation more important than the other stakeholder groups, which may be a result of a more analytical knowledge of the underlying elements of fire occurrence. Differences between respondents may also be a result of practical experiences. A study by Costanza and Moody (2011) indicated that participation in firefighting and fire suppression can prioritize weather conditions, such as wind, humidity, and fuel moisture, which stems from their direct experience with fire suppression. It emphasizes a shift towards considering factors that primarily affect fire behavior, including the rate of spread and the total area burned.

Representatives of action forces tended to give higher ranking to fuel quality, fuel load, and atmospheric stability, especially in the future. While the use of fire as a management tool is commonly implemented in other regions (Ascoli and Bovio, 2013; Fernandes, 2013), in the Alpine region, it raises concerns about smoke emissions and an increased fire risk in surrounding areas (Stephens and Moghaddas, 2005). In some parts of France and Italy, prescribed burning is a well understood and used management technique to control the fire regime (Ascoli and Bovio, 2013; Fernandes, 2013), but it is forbidden in other Alpine countries like Austria, Germany, and Switzerland (Montiel-Molina, 2013; Müller et al., 2020).

On the socio-economic side, respondents' emphasis on recreational activities indicates that human use of forest landscapes remains a primary ignition context, consistent with prior work linking ignition patterns to human activities and access (Conedera et al., 2018; Ganteaume et al., 2013; Martínez et al., 2009; Sande Silva et al., 2010). This perception contrasts with evidence from the Brazilian Amazon, where ignitions are described as largely tied to forest conversion and agricultural maintenance. These practices include the use of fire to eliminate biomass after deforestation (known as slash-and-burn), as well as to renew vegetation and control weeds during the dry season. Such fires can occasionally spread into adjacent forests, leading to wildfires (Silveira et al., 2020).

The importance of the different socio-economic factors depends on the socio-economic differences in the studied countries. For example, rural abandonment is considered the main risk factor for major forest fires in France and Italy, while recreational activities are the most important socio-economic driver of forest fires in Austria, Germany and Switzerland (Müller et al., 2020).

In our study, all three stakeholder groups expected that the importance of rural abandonment would increase, indicating a shared concern about land-use change as an emerging forest fire driver. Rural abandonment implies loss of pastures, forest regrowth, and replacement of cultural firebreaks with fire-prone scrublands, resulting in an increase in the amount and connectivity of fuel (Müller et al., 2020).

Scientists also ranked public awareness with a higher importance than authorities. In a similar context Raftoyannis et al. (2014) highlighted differences in the perceptions of the effectiveness of fire prevention and suppression measures between forestry officers and researchers in the European Mediterranean region. Forestry officers believed that public awareness is less important than fire suppression measures. This clearly indicates the need of raising awareness among authorities and forestry officers about the role of dissemination and communication measures in the future.

4.4. From drivers to policy levers: Translating perceived risk into governance-relevant action

The findings of our study allow us to translate the perceived drivers for fire ignitions into three governance-relevant policy levers and the related implications within cross-sector wildfire risk governance framings.

4.4.1. Reducing avoidable human ignitions

It was found that recreational activities were consistently rated as an important driver across all studied stakeholder groups, indicating convergence on visitor-related ignition contexts as a priority intervention point. This aligns with the scientific evidences that human activities and access can shape ignition patterns and fire occurrence in European settings (Benefield and Chen, 2022; Devenport and Edgeley, 2025). This points to a more prevention oriented strategy on “everyday ignition reduction” with a clear behavioral guidance for residents and visitors, targeted restrictions where risk peaks, proportionate enforcement, and consistent risk communication are important (EASAC, 2025).

4.4.2. Firewise landscape management

Stakeholders, particularly action forces, assigned high importance to fuel-related drivers, consistent with an operational focus on conditions shaping fire behavior and the potential for larger burned areas (Bekar et al., 2020; Costanza and Moody, 2011).

Rural abandonment was perceived as increasingly important driver for future wildfire risk, and the revised discussion specifies a plausible mechanism whereby abandonment can increase fuel accumulation and connectivity via regrowth and the loss of “cultural firebreaks” (Müller et al., 2020). This portfolio logic is consistent with European science–policy emphasis on firewise landscape management and sustainable private land management as components of an integrated prevention strategy (EASAC, 2025). It also aligns with OECD's call to mainstream wildfire prevention across policy frameworks, positioning land management and forestry strategies as central vehicles for operationalizing fuel-related risk reduction (OECD, 2023).

4.4.3. Governing exposure and strengthening capacity

Variation in how groups weight “public awareness” and governance-related factors is policy-relevant because it signals potential coordination gaps: if some actors undervalue awareness-raising while others see it as central, prevention efforts may remain fragmented or more suppression oriented. Governance levers therefore include coordination mechanisms linking land management, civil protection, and the recreation and tourism sector to reduce ignition pressure and integrate wildfire risk assessment into spatial and landscape planning. This will help to minimize exposure of natural environments and reduce vulnerability of threatened settlements, and provide means for a sustained education and risk communication to build fire literacy among residents, landowners, and visitors (EASAC, 2025).

The perception patterns identified in this study sit within an evolving common-policy framework on integrated fire management that has

emerged at the international and European levels in recent years. The Global Fire Management Hub, which was launched by the Food and Agriculture Organization (FAO) during the 8th International Wildland Fire Conference in 2023, has become the leading international partnership for countries to enhance their institutional and operational capabilities in integrated fire management (FAO, 2024). At the European level, recent science–policy advice has called for an integrated EU framework for landscape fire-risk governance grounded in stakeholder collaboration and vertical and horizontal policy harmonization (EASAC, 2025).

Overall, translating the perceived drivers into policy levers yields a coherent risk-reduction portfolio combining (i) visitor- and negligence-oriented ignition prevention, (ii) measures reducing fuel build-up and connectivity under land-use change and abandonment dynamics, and (iii) cross-sector governance mechanisms that embed wildfire risk in planning and strengthen communication and preparedness.

5. Conclusion

This study presents the differences in the perceptions of selected stakeholders regarding the trends and driving factors of forest fires in the countries of the Alpine region. Our findings revealed a significant difference in perceptions of three stakeholder groups regarding the number of forest fires and burnt areas. Precipitation, temperature, and wind were chosen by all stakeholders to be the most important natural factors, whereas recreational activities, rural abandonment, and owner awareness were the most important socio-economic drivers of the past and current fire regime. These findings identify and understand the similarities and differences in the stakeholders' perceptions of the current and future fire regime, which may lead to varying fire management priorities and actions affecting the forest fire situation in the Alps. Therefore, an intensified education, cross-sectoral dialogue, collaboration, and mutual understanding are important to obtain a common understanding of existing facts and different perceptions among all groups of stakeholders. More efforts in research and transdisciplinary exchange are needed to enhance effective wildfire preparedness and response strategies in the Alpine region.

Although a limitation of the study was the non-representative set of stakeholders participating in the survey with the snowball sampling strategy it was possible to capture the main aspects that influence the current fire regime. Participation was shaped by the networks of the contacted seed actors and by respondents' self-motivation and interests, which resulted in uneven participation across Alpine countries. Accordingly, the study does not support cross-country inference, and results are interpreted as patterns among surveyed stakeholder types. In addition, the findings of this study relied on data originating from a survey of the year 2019, which reflects the stakeholder perceptions from representatives of countries in the Alps region from the past. In linking those findings with the current fire management policies in Europe it became evident, how timely the results are, and that there are still a lot of efforts needed to implement such integrated fire management strategies. Given the increasing frequency of extreme fire seasons and the changing of policies, future research should employ data collection methods that integrate multiple sources and include a greater number of participants to enable the tracking of shifting perceptions and allow for the evaluation of the effectiveness of combined research, education, and dialogue initiatives in integrated forest fire management in the Alps regions.

CRediT authorship contribution statement

The Trung Hoang: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Lena Vilà-Vilardell:** Writing – review & editing, Resources, Investigation. **Mortimer M. Müller:** Writing – review & editing, Resources, Investigation. **Giorgio Vacchiano:** Writing – review & editing. **Gianni Boris Pezzatti:** Writing – review & editing, Visualization. **Sylvain Dupire:** Writing – review & editing, Visualization. **Christian Schunk:** Writing – review & editing. **Harald Vacik:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Acknowledgements

This work was partly funded by the project “Forest fires in the Alps - state of knowledge and future challenges” in the context of the Alpine Space Project AlpGov and the EUSALP (Action Group 8) of the European Regional Development Fund and the Ministry of Agriculture and Forestry, Climate and Environmental Protection, Regions and Water Management in Austria as well as by the Austrian Forest Fire Research II at the Institute of Silviculture, Department for Ecosystem Management, Climate and Biodiversity, BOKU University, Vienna.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.forpol.2026.103830.

Data availability

Data will be made available on request.

References

Aguilar, S., Montiel, C., 2011. The challenge of applying governance and sustainable development to wildland fire management in southern Europe. J. For. Res. 22, 627–639. https://doi.org/10.1007/s11676-011-0168-6.

Alló, M., Loureiro, M.L., 2020. Assessing preferences for wildfire prevention policies in Spain. Forest Policy Econ. 115, 102145. https://doi.org/10.1016/j.forpol.2020.102145.

Arpaci, A., Eastaugh, C.S., Vacik, H., 2013. Selecting the best performing fire weather indices for Austrian ecoregions. Theor. Appl. Climatol. 114, 393–406. https://doi.org/10.1007/s00704-013-0839-7.

Ascoli, D., Bovio, G., 2013. Prescribed burning in Italy: issues, advances and challenges. IForest - Biogeosci. For. 6, 79. https://doi.org/10.3832/ifor0803-006.

Auer, I., Böhm, R., Jurkovic, A., Lipa, W., Orlik, A., Potzmann, R., Schöner, W., Ungersböck, M., Matulla, C., Briffa, K., Jones, P., Efthymiadis, D., Brunetti, M., Nanni, T., Maugeri, M., Mercalli, L., Mestre, O., Moisselin, J.-M., Begert, M., Müller-Westermeier, G., Kveton, V., Bochnicek, O., Stastny, P., Lapin, M., Szalai, S., Szentimrey, T., Cegnar, T., Dolinar, M., Gajic-Capka, M., Zaninovic, K., Majstorovic, Z., Nieplova, E., 2007. HISTALP—historical instrumental climatological surface time series of the greater Alpine region. Int. J. Climatol. 27, 17–46. https://doi.org/10.1002/joc.1377.

Bekar, İ., Tavşanoğlu, Ç., Pezzatti, G.B., Vacik, H., Pausas, J.G., Bugmann, H., Petter, G., 2020. Cross-regional modelling of fire occurrence in the Alps and the Mediterranean Basin. Int. J. Wildland Fire 29, 712–722. https://doi.org/10.1071/WF19158.

Benefield, A., Chen, J., 2022. Examining the influence of outdoor recreation on anthropogenic wildfire regime of the southern Rocky Mountains. Nat. Hazards 111, 523–545. https://doi.org/10.1007/s11069-021-05065-1.

Canepa, A., Drogo, F., 2021. Wildfire crime, apprehension and social vulnerability in Italy. Forest Policy Econ. 122, 102330. https://doi.org/10.1016/j.forpol.2020.102330.

Carvalho, A., Flannigan, M., Logan, K., Johnston, L., Miranda, A., Borrego, C., 2010. The impact of spatial resolution on area burned and fire occurrence projections in Portugal under climate change. Clim. Chang. 98, 177–197. https://doi.org/10.1007/s10584-009-9667-2.

Castillo, J.A., Smith-Ramírez, C., Claramunt, V., 2021. Differences in stakeholder perceptions about native forest: implications for developing a restoration program. Restor. Ecol. 29, e13293. https://doi.org/10.1111/rec.13293.

Conedera, M., Cesti, G., Pezzatti, G.B., Zumbrunnen, T., Spinedi, F., 2006. Lightning-induced fires in the Alpine region: an increasing problem. For. Ecol. Manag. 234, S68. https://doi.org/10.1016/j.foreco.2006.08.096.

Conedera, M., Krebs, P., Valese, E., Cocca, G., Schunk, C., Menzel, A., Vacik, H., Cane, D., Japelj, A., Muri, B., Ricotta, C., Oliveri, S., Pezzatti, G.B., 2018. Characterizing Alpine pyrogeography from fire statistics. Appl. Geogr. 98, 87–99. https://doi.org/10.1016/j.apgeog.2018.07.011.

Costa, R., Menichini, T., 2013. A multidimensional approach for CSR assessment: the importance of the stakeholder perception. Expert Syst. Appl. 40, 150–161. https://doi.org/10.1016/j.eswa.2012.07.028.

Costanza, J.K., Moody, A., 2011. Deciding where to burn: stakeholder priorities for prescribed burning of a fire-dependent ecosystem. Ecol. Soc. 16, art14. https://doi.org/10.5751/ES-03897-160114.
Csontos, P., Cseresnyés, I., 2015. Fire-risk evaluation of Austrian pine stands in Hungary – effects of drought conditions and slope aspect on fire spread and fire behavior. Carpathian J. Earth Environ. Sci. 10 (3), 247–254.
Curt, T., Fréjaville, T., Lahaye, S., Curt, T., Fréjaville, T., Lahaye, S., 2016. Modelling the spatial patterns of ignition causes and fire regime features in southern France: implications for fire prevention policy. Int. J. Wildland Fire 25, 785–796. https://doi.org/10.1071/WF15205.
Devenport, S.E., Edgeley, C.M., 2025. Visitor preferences and human-caused wildfire prevention: a survey of public land users in the US southwest. J. For. 123, 171–196. https://doi.org/10.1007/s44392-024-00003-y.
Dunn, O.J., 1964. Multiple comparisons using rank sums. Technometrics 6, 241–252. https://doi.org/10.1080/00401706.1964.10490181.
Dupire, S., Curt, T., Bigot, S., 2017. Spatio-temporal trends in fire weather in the French Alps. Sci. Total Environ. 595, 801–817. https://doi.org/10.1016/j.scitotenv.2017.04.027.
Dupire, S., Curt, T., Bigot, S., Fréjaville, T., 2019. Vulnerability of forest ecosystems to fire in the French Alps. Eur. J. For. Res. 138, 813–830. https://doi.org/10.1007/s10342-019-01206-1.
Dwivedi, P., Jagadish, A., Schelhas, J., 2016. Perceptions of stakeholder groups about the participation of African American family Forest landowners in Federal Landowner Assistance Programs. J. For. 114, 89–96. https://doi.org/10.5849/jof.14-152.
EASAC – the European Academies'' Science Advisory Council, 2025. Changing wildfires: policy options for a fire-literate and fire-adapted Europe. Verlag der Österreichischen Akademie der Wissenschaften. https://doi.org/10.1553/EASAC_Report_Changing-Wildfires_2025.
FAO, 2024. Integrated fire management voluntary guidelines – Principles and strategic actions. Forestry Working Paper, No. 41, Second edition. https://doi.org/10.4060/cd1090en.
Fernandes, P.M., 2013. Fire-smart management of forest landscapes in the Mediterranean basin under global change. Landsc. Urban Plan. 110, 175–182. https://doi.org/10.1016/j.landurbplan.2012.10.014.
Fernandes, P.M., Botelho, H.S., 2003. A review of prescribed burning effectiveness in fire hazard reduction. Int. J. Wildland Fire 12, 117–128. https://doi.org/10.1071/wf02042.
Fernandez-Anez, N., Krasovskiy, A., Müller, M., Vacik, H., Baetens, J., Hukić, E., Kapovic Solomun, M., Atanassova, I., Glushkova, M., Bogunović, I., Fajković, H., Djuma, H., Boustras, G., Adámek, M., Devetter, M., Hrabalikova, M., Huska, D., Martínez Barroso, P., Vaverková, M.D., Zumr, D., Jõgiste, K., Metslaid, M., Koster, K., Köster, E., Pumpanen, J., Ribeiro-Kumara, C., Di Prima, S., Pastor, A., Rumpel, C., Seeger, M., Daliakopoulos, I., Daskalakou, E., Koutroulis, A., Papadopoulou, M.P., Stampoulidis, K., Xanthopoulos, G., Aszalós, R., Balázs, D., Kertész, M., Valkó, O., Finger, D.C., Thorsteinsson, T., Till, J., Bajocco, S., Gelsomino, A., Amodio, A.M., Novara, A., Salvati, L., Telesca, L., Ursino, N., Jansons, A., Kitenberga, M., Stivrins, N., Brazaitis, G., Marozas, V., Cojocaru, O., Gumeniuc, I., Sfecla, V., Imeson, A., Veraverbeke, S., Mikalsen, R.F., Koda, E., Osinski, P., Castro, A.C.M., Nunes, J.P., Oom, D., Vieira, D., Rusu, T., Bojović, S., Djordjevic, D., Popovic, Z., Protic, M., Sakan, S., Glasa, J., Kacikova, D., Lichner, L., Majlingova, A., Vido, J., Ferk, M., Tičar, J., Zorn, M., Zupanc, V., Hinojosa, M.B., Knicker, H., Lucas-Borja, M. E., Pausas, J., Prat-Guitart, N., Ubeda, X., Vilar, L., Destouni, G., Ghajarnia, N., Kalantari, Z., Seifollahi-Aghmiuni, S., Dindaroglu, T., Yakupoglu, T., Smith, T., Doerr, S., Cerda, A., 2021. Current wildland fire patterns and challenges in Europe: a synthesis of National Perspectives. Air Soil Water Res. 14, 11786221211028185. https://doi.org/10.1177/11786221211028185.
Galiana, L., Aguilar, S., Lázaro, A., 2013. An assessment of the effects of forest-related policies upon wildland fires in the European Union: applying the subsidiarity principle. In: For. Policy Econ. The FIRE PARADOX Project: Setting the Basis for a Shift in the Forest Fire Policies in Europe, 29, pp. 36–44. https://doi.org/10.1016/j.forpol.2012.10.010.
Ganteaume, A., Camia, A., Jappiot, M., et al., 2013. A review of the main driving factors of forest fire ignition over Europe. Environ. Manag. 51, 651–662. https://doi.org/10.1007/s00267-012-9961-z.
Gerberding, K., Schirpke, U., 2025. Mapping the probability of forest fire hazard across the European Alps under climate change scenarios. J. Environ. Manag. 377, 124600. https://doi.org/10.1016/j.jenvman.2025.124600.
Gobiet, A., Kotlarski, S., Beniston, M., Heinrich, G., Rajczak, J., Stoffel, M., 2014. 21st century climate change in the European Alps—a review. Sci. Total Environ. 493, 1138–1151. https://doi.org/10.1016/j.scitotenv.2013.07.050.
Grilli, G., Jonkisz, J., Ciolli, M., Lesinski, J., 2016. Mixed forests and ecosystem services: investigating stakeholders' perceptions in a case study in the polish Carpathians. Forest Policy Econ. 66, 11–17. https://doi.org/10.1016/j.forpol.2016.02.003.
Hickey, G.M., Innes, J.L., 2005. Monitoring sustainable Forest Management in Different Jurisdictions. Environ. Monit. Assess. 108, 241–260. https://doi.org/10.1007/s10661-005-4329-y.
Hochberg, Y., Tamhane, A.C., 1987. Multiple Comparison Procedures. John Wiley & Sons. https://doi.org/10.1002/9780470316672.
Huppmann, D., Keiler, M., Riahi, K., Rieder, H., Brad, A., Formayer, H., Fuchs, S., Kettner, C., Kranzl, L., Novy, A., Resch, G., Schinko, T., Schmidt, J., Schöniger, F., Steininger, K., Spittler, N., Donner, A., Holzinger, M., Hummel, E.-M., Mayer, M., Rauchenwald, V., Scheifinger, K., 2025. Summary for policymakers. In: Huppmann, D., Keiler, M., Riahi, K., Rieder, H. (Eds.), Second Austrian Assessment Report on Climate Change (AAR2) of the Austrian Panel on Climate Change (APCC). Austrian Academy of Sciences Press, Vienna. https://doi.org/10.1553/aar2-spm.
Kahneman, D., Tversky, A., 1979. Prospect theory: an analysis of decision under risk. Econometrica 47, 263. https://doi.org/10.2307/1914185.
Kotlarski, S., Gobiet, A., Morin, S., Olefs, M., Rajczak, J., Samacoïts, R., 2023. 21st century alpine climate change. Clim. Dyn. 60, 65–86. https://doi.org/10.1007/s00382-022-06303-3.
Kruskal, W.H., Wallis, W.A., 1952. Use of ranks in one-criterion variance analysis. J. Am. Stat. Assoc. 47, 583–621. https://doi.org/10.1080/01621459.1952.10483441.
Lahaye, S., Curt, T., Fréjaville, T., Sharples, J., Paradis, L., Hély, C., 2018. What are the drivers of dangerous fires in Mediterranean France? Int. J. Wildland Fire 27, 155. https://doi.org/10.1071/WF17087.
Maguire, L.A., Albright, E.A., 2005. Can behavioral decision theory explain risk-averse fire management decisions?. In: For. Ecol. Manag., Relative Risk Assessments for Decision –making Related to Uncharacteristic Wildfire, 211, pp. 47–58. https://doi.org/10.1016/j.foreco.2005.01.027.
Martínez, J., Vega-Garcia, C., Chuvieco, E., 2009. Human-caused wildfire risk rating for prevention planning in Spain. J. Environ. Manag. 90 (2), 1241–1252. https://doi.org/10.1016/j.jenvman.2008.07.005.
Merschel, A.G., Spies, T.A., Heyerdahl, E.K., 2014. Mixed-conifer forests of Central Oregon: effects of logging and fire exclusion vary with environment. Ecol. Appl. 24, 1670–1688. https://doi.org/10.1890/13-1585.1.
Molina-Terrén, D.M., Cardil, A., Kobziar, L.N., 2016. Practitioner perceptions of wildland fire management across South Europe and Latin America. Forests 7, 184. https://doi.org/10.3390/f7090184.
Montiel-Molina, C., 2013. Comparative assessment of wildland fire legislation and policies in the European Union: towards a fire framework directive. In: For. Policy econ., the FIRE PARADOX Project: Setting the Basis for a Shift in the Forest Fire Policies in Europe, 29, pp. 1–6. https://doi.org/10.1016/j.forpol.2012.11.006.
Müller, M.M., Vacik, H., 2017. Characteristics of lightnings igniting forest fires in Austria. Agric. For. Meteorol. 240–241, 26–34. https://doi.org/10.1016/j.agrformet.2017.03.020.
Müller, M.M., Vacik, H., Diendorfer, G., Arpaci, A., Formayer, H., Gossow, H., 2013. Analysis of lightning-induced forest fires in Austria. Theor. Appl. Climatol. 111, 183–193. https://doi.org/10.1007/s00704-012-0653-7.
Müller, M.M., Vacik, H., Valese, E., 2015. Anomalies of the Austrian Forest fire regime in comparison with other Alpine countries: a research note. Forests 6, 903–913. https://doi.org/10.3390/f6040903.
Müller, Vilardell L.V., Vacik, H., Mayer, C., Mayr, S., Carrega, P., Duche, Y., Lahaye, Sebastatien, Böttcher, F., Maier, H., Schunk, C., Zimmermann, L., Ascoli, D., Cotterchio, A., Fiorucci, P., Gottero, F., Pirone, S., Rizzolo, R., Vacchiano, G., Valese, E., Jereb, L., Kolsek, M., Kosicek, Bostjan, Papez, Joze, Conedera, M., Ghiringhelli, A., Pezzatti, G.B., Sautter, M., 2020. FOREST FIRES IN THE ALPS - State of Knowledge, Future Challenges and Options for an Integrated Fire Management - White Paper for Policy Makers. https://doi.org/10.13140/RG.2.2.15609.42081.
Müller, M.M., Vilà-Vilardell, L., Vacik, H., 2020. Towards an integrated forest fire danger assessment system for the European Alps. Ecol. Inform. 60, 101151. https://doi.org/10.1016/j.ecoinf.2020.101151.
Naficy, C., Sala, A., Keeling, E.G., Graham, J., DeLuca, T.H., 2010. Interactive effects of historical logging and fire exclusion on ponderosa pine forest structure in the northern Rockies. Ecol. Appl. 20, 1851–1864. https://doi.org/10.1890/09-0217.1.
OECD, 2023. Taming wildfires in the context of climate change. OECD Publishing. https://doi.org/10.1787/dd00c367-en.
Paletto, A., Maino, F., De Meo, I., Ferretti, F., 2013. Perception of Forest values in the Alpine Community of Trentino Region (Italy). Environ. Manag. 51, 414–422. https://doi.org/10.1007/s00267-012-9974-7.
Pandey, P., Huidobro, G., Lopes, L.F., Ganteaume, A., Ascoli, D., Colaco, C., Xanthopoulos, G., Giannaros, T.M., Gazzard, R., Boustras, G., Steelman, T., Charlton, V., Ferguson, E., Kirschner, J., Little, K., Stoof, C., Nikolakis, W., Fernández-Blanco, C.R., Ribotta, C., Lambrechts, H., Fernandez, M., Dossi, S., 2023. A global outlook on increasing wildfire risk: current policy situation and future pathways. Trees For. People 14, 100431. https://doi.org/10.1016/j.tfp.2023.100431.
Papathoma-Köhle, M., Hausharter, D., Schlögl, M., Fuchs, S., 2025. Towards a wildfire vulnerability index using expert judgement. Int. J. Wildland Fire 34. https://doi.org/10.1071/WF24114.
Pereira, P., Mierauskas, P., Novara, A., 2016. Stakeholders' perceptions about fire impacts on Lithuanian protected areas. Land Degrad. Dev. 27, 871–883. https://doi.org/10.1002/ldr.2290.
Pezzatti, G.B., Zumbrunnen, T., Bürgi, M., Ambrosetti, P., Conedera, M., 2013. Fire regime shifts as a consequence of fire policy and socio-economic development: an analysis based on the change point approach. In: For. Policy Econ. The FIRE PARADOX Project: Setting the Basis for a Shift in the Forest fire Policies in Europe, 29, pp. 7–18. https://doi.org/10.1016/j.forpol.2011.07.002.
R Core Team, 2024. R: A Language and Environment for Statistical Computing.
Raftoyannis, Y., Nocentini, S., Marchi, E., Calama Sainz, R., Garcia Guemes, C., Pilas, I., Peric, S., Amaral Paulo, J., Moreira-Marcelino Ana, C., Costa-Ferreira, M., Kakouris, E., Lindner, M., 2014. Perceptions of forest experts on climate change and fire management in European Mediterranean forests. IForest - Biogeosci. For. 7, 33–41. https://doi.org/10.3832/ifor0817-006.
Rego, F., Louro, G., Constantino, L., 2013. The impact of changing wildfire regimes on wood availability from Portuguese forests. In: For. Policy Econ. The FIRE PARADOX Project: Setting the Basis for a Shift in the Forest Fire Policies in Europe, 29, pp. 56–61. https://doi.org/10.1016/j.forpol.2012.11.010.
Rizzolo, R., 2016. Fuel Models Development to Support Spatially-Explicit Forest Fire Modelling in Eastern Italian Alps [WWW Document]. URL. http://paduaresearch.cab.unipd.it/9265/. accessed 6.14.21).

Sande Silva, J., Rego, F., Fernandes, P., Rigolot, E. (editors), 2010. Towards Integrated Fire Management – Outcomes of the European Project Fire Paradox. Research Report 23. European Forest Institute.

Schunk, C., Wastl, C., Leuchner, M., Schuster, C., Menzel, A., 2013. Forest fire danger rating in complex topography – results from a case study in the Bavarian Alps in autumn 2011. Nat. Hazards Earth Syst. Sci. 13, 2157–2167. https://doi.org/10.5194/nhess-13-2157-2013.

Sevinc, V., Kucuk, O., Goltas, M., 2020. A Bayesian network model for prediction and analysis of possible forest fire causes. For. Ecol. Manag. 457, 117723. https://doi.org/10.1016/j.foreco.2019.117723.

Silveira, M.V.F., Petri, C.A., Broggio, I.S., Chagas, G.O., Macul, M.S., Leite, C.C.S.S., Ferrari, E.M.M., Amim, C.G.V., Freitas, A.L.R., Motta, A.Z.V., Carvalho, L.M.E., Silva Junior, C.H.L., Anderson, L.O., Aragão, L.E.O.C., 2020. Drivers of fire anomalies in the Brazilian Amazon: lessons learned from the 2019 fire crisis. Land 9, 516. https://doi.org/10.3390/land9120516.

Slovic, P., 1987. Perception of risk. Science 236, 280–285. https://doi.org/10.1126/science.3563507.

Slovic, P., Fischhoff, B., Lichtenstein, S., 1982. Why study risk perception? Risk Anal. 2, 83–93. https://doi.org/10.1111/j.1539-6924.1982.tb01369.x.

Stephens, S.L., Moghaddas, J.J., 2005. Experimental fuel treatment impacts on forest structure, potential fire behavior, and predicted tree mortality in a California mixed conifer forest. For. Ecol. Manag. 215, 21–36. https://doi.org/10.1016/j.foreco.2005.03.070.

Swim, J.K., Stern, P.C., Doherty, T.J., Clayton, S., Reser, J.P., Weber, E.U., Gifford, R., Howard, G.S., 2011. Psychology's contributions to understanding and addressing global climate change. Am. Psychol. 66, 241–250. https://doi.org/10.1037/a0023220.

Tedim, F., Leone, V., Amraoui, M., Bouillon, C., Coughlan, M.R., Delogu, G.M., Fernandes, P.M., Ferreira, C., McCaffrey, S., McGee, T.K., Parente, J., Paton, D., Pereira, M.G., Ribeiro, L.M., Viegas, D.X., Xanthopoulos, G., 2018. Defining extreme wildfire events: difficulties, challenges, and impacts. Fire 1, 9. https://doi.org/10.3390/fire1010009.

Tedim, F., Leone, V., Lovreglio, R., Xanthopoulos, G., Chas-Amil, M.-L., Ganteaume, A., Efe, R., Royé, D., Fuerst-Bjeliš, B., Nikolov, N., Musa, S., Milenković, M., Correia, F., Conedera, M., Boris Pezzatti, G., 2022. Forest fire causes and motivations in the southern and South-Eastern Europe through experts' perception and applications to current policies. Forests 13, 562. https://doi.org/10.3390/f13040562.

Trnka, M., Balek, J., Stepanek, P., Zahradníček, P., Mozny, M., Eitzinger, J., Zalud, Z., Formayer, H., Turna, M., Nejedlik, P., Semeradova, D., Hlavinka, P., Brázdil, R., 2016. Drought trends over part of Central Europe between 1961 and 2014. Clim. Res. 70. https://doi.org/10.3354/cr01420.

Turco, M., Rosa-Cánovas, J.J., Bedia, J., Jerez, S., Montávez, J.P., Llasat, M.C., Provenzale, A., 2018. Exacerbated fires in Mediterranean Europe due to anthropogenic warming projected with non-stationary climate-fire models. Nat. Commun. 9, 3821. https://doi.org/10.1038/s41467-018-06358-z.

Vacchiano, G., Foderi, C., Berretti, R., Marchi, E., Motta, R., 2018. Modeling anthropogenic and natural fire ignitions in an inner-alpine valley. Nat. Hazards Earth Syst. Sci. 18, 935–948. https://doi.org/10.5194/nhess-18-935-2018.

Vacik, H., Arndt, N., Arpaci, A., Koch, V., Müller, M.M., Gossow, H., 2011. Characterisation of forest fires in Austria. Austrian J. For. Sci. 128, 1–31.

Valese, E., Conedera, M., Held, A.C., Ascoli, D., 2014. Fire, humans and landscape in the European Alpine region during the Holocene. Anthropocene 6, 63–74. https://doi.org/10.1016/j.ancene.2014.06.006.

Wastl, C., Schunk, C., Leuchner, M., Pezzatti, G.B., Menzel, A., 2012. Recent climate change: long-term trends in meteorological forest fire danger in the Alps. Agric. For. Meteorol. 1–13. https://doi.org/10.1016/j.agrformet.2012.04.001.

Wastl, C., Schunk, C., Lüpke, M., Cocca, G., Conedera, M., Valese, E., Menzel, A., 2013. Large-scale weather types, forest fire danger, and wildfire occurrence in the Alps. Agric. For. Meteorol. 168, 15–25. https://doi.org/10.1016/j.agrformet.2012.08.011.

Wunder, S., Calkin, D.E., Charlton, V., Feder, S., Martínez de Arano, I., Moore, P., Rodríguez y Silva, F., Tacconi, L., Vega-García, C., 2021. Resilient landscapes to prevent catastrophic forest fires: socioeconomic insights towards a new paradigm. Forest Policy Econ. 128, 102458. https://doi.org/10.1016/j.forpol.2021.102458.

Zumbrunnen, T., Menéndez, P., Bugmann, H., Conedera, M., Gimmi, U., Bürgi, M., 2012. Human impacts on fire occurrence: a case study of hundred years of forest fires in a dry alpine valley in Switzerland. Reg. Environ. Chang. 12, 935–949. https://doi.org/10.1007/s10113-012-0307-4.